

## SAFETY DATA SHEET

Creation Date 13-August-2018

Revision Date 30-March-2024

Revision Number 4

### 1. Identification

**Product Name** 4-Aminostyrene

**Cat No. :** L11845

**CAS-No** 1520-21-4  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                      |            |
|------------------------------------------------------|------------|
| Acute oral toxicity                                  | Category 4 |
| Acute dermal toxicity                                | Category 4 |
| Acute Inhalation Toxicity                            | Category 4 |
| Skin Corrosion/Irritation                            | Category 2 |
| Serious Eye Damage/Eye Irritation                    | Category 2 |
| Respiratory Sensitization                            | Category 1 |
| Carcinogenicity                                      | Category 2 |
| Specific target organ toxicity (single exposure)     | Category 3 |
| Target Organs - Respiratory system.                  |            |
| Specific target organ toxicity - (repeated exposure) | Category 2 |
| Target Organs - Blood, Hematopoietic System.         |            |

#### Label Elements

**Signal Word**

Danger

**Hazard Statements**

Harmful if swallowed, in contact with skin or if inhaled  
Causes skin irritation  
Causes serious eye irritation  
May cause allergy or asthma symptoms or breathing difficulties if inhaled  
May cause respiratory irritation  
Suspected of causing cancer  
May cause damage to organs through prolonged or repeated exposure  
Harmful if inhaled

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Wear protective gloves/protective clothing/eye protection/face protection  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear respiratory protection

**Response**

IF exposed or concerned: Get medical advice/attention  
IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Call a POISON CENTER/ doctor if you feel unwell  
Rinse mouth  
If experiencing respiratory symptoms: Call a POISON CENTER/doctor  
Take off contaminated clothing and wash it before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component               | CAS-No    | Weight % |
|-------------------------|-----------|----------|
| Benzenamine, 4-ethenyl- | 1520-21-4 | <=100    |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

|                                        |                                                                                                                                                                                                                                                                  |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                                                |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                                                     |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water.                                                                                                                                                                                                     |
| <b>Most important symptoms/effects</b> | May cause allergy or asthma symptoms or breathing difficulties if inhaled. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                            |

## 5. Fire-fighting measures

|                                         |                                                                                                                                                                      |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Powder. Water spray. In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                             |
| <b>Flash Point</b>                      | 99 °C / 210.2 °F                                                                                                                                                     |
| <b>Method -</b>                         | No information available                                                                                                                                             |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                                             |
| <b>Explosion Limits</b>                 |                                                                                                                                                                      |
| <b>Upper</b>                            | No data available                                                                                                                                                    |
| <b>Lower</b>                            | No data available                                                                                                                                                    |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                             |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                             |

### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
-

## 6. Accidental release measures

|                                  |                                                                             |
|----------------------------------|-----------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required. |
| <b>Environmental Precautions</b> | Should not be released into the environment.                                |

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

|                 |                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid ingestion and inhalation. |
| <b>Storage.</b> | Keep refrigerated. Keep containers tightly closed in a cool, well-ventilated place. Protect from direct sunlight. Incompatible Materials. Oxidizing agent.    |

## 8. Exposure controls / personal protection

### Exposure Guidelines

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Goggles

#### **Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** conforming to EN14387 Organic gases and vapours filter Type A Brown

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                         |                             |
|-----------------------------------------|-----------------------------|
| <b>Physical State</b>                   | Low melting solid           |
| <b>Appearance</b>                       | No information available    |
| <b>Odor</b>                             | Odorless                    |
| <b>Odor Threshold</b>                   | No information available    |
| <b>pH</b>                               | No information available    |
| <b>Melting Point/Range</b>              | 23 - 24 °C / 73.4 - 75.2 °F |
| <b>Boiling Point/Range</b>              | 249 °C / 480 °F @ 760 mmHg  |
| <b>Flash Point</b>                      | 99 °C / 210.2 °F            |
| <b>Evaporation Rate</b>                 | No information available    |
| <b>Flammability (solid,gas)</b>         | Not applicable              |
| <b>Flammability or explosive limits</b> |                             |
| Upper                                   | No data available           |

|                                        |                          |
|----------------------------------------|--------------------------|
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Specific Gravity                       | 1.014 g/cm3              |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |
| Molecular Formula                      | C8 H9 N                  |
| Molecular Weight                       | 119.17                   |

## 10. Stability and reactivity

|                                  |                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                  |
| Stability                        | Light sensitive. Stable under normal conditions.                                            |
| Conditions to Avoid              | Incompatible products.                                                                      |
| Incompatible Materials           | Oxidizing agent                                                                             |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Nitrogen oxides (NO <sub>x</sub> ) |
| Hazardous Polymerization         | Hazardous polymerization may occur upon depletion of inhibitor.                             |
| Hazardous Reactions              | None under normal processing.                                                               |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | May cause sensitization by inhalation                                                    |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component               | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------|-----------|------------|------------|------------|------------|------------|
| Benzenamine, 4-ethenyl- | 1520-21-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                   |                          |
|-------------------|--------------------------|
| Mutagenic Effects | No information available |
|-------------------|--------------------------|

|                      |                           |
|----------------------|---------------------------|
| Reproductive Effects | No information available. |
|----------------------|---------------------------|

|                       |                           |
|-----------------------|---------------------------|
| Developmental Effects | No information available. |
|-----------------------|---------------------------|

|                |                           |
|----------------|---------------------------|
| Teratogenicity | No information available. |
|----------------|---------------------------|

|                          |                            |
|--------------------------|----------------------------|
| STOT - single exposure   | Respiratory system         |
| STOT - repeated exposure | Blood Hematopoietic System |

|                   |                          |
|-------------------|--------------------------|
| Aspiration hazard | No information available |
|-------------------|--------------------------|

|                                    |                                                                                                |
|------------------------------------|------------------------------------------------------------------------------------------------|
| Symptoms / effects, both acute and | Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling |
|------------------------------------|------------------------------------------------------------------------------------------------|

|                                        |                                                                                        |
|----------------------------------------|----------------------------------------------------------------------------------------|
| <b>delayed</b>                         | of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Endocrine Disruptor Information</b> | No information available                                                               |
| <b>Other Adverse Effects</b>           | The toxicological properties have not been fully investigated.                         |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

|                                      |                           |
|--------------------------------------|---------------------------|
| <b>Persistence and Degradability</b> | No information available  |
| <b>Bioaccumulation/ Accumulation</b> | No information available. |
| <b>Mobility</b>                      | No information available. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                             |                               |
|-----------------------------|-------------------------------|
| <b>UN-No</b>                | UN2811                        |
| <b>Proper Shipping Name</b> | TOXIC SOLIDS, ORGANIC, N.O.S. |
| <b>Technical Name</b>       | Benzenamine, 4-ethenyl-       |
| <b>Hazard Class</b>         | 6.1                           |
| <b>Packing Group</b>        | III                           |

### TDG

|                             |                              |
|-----------------------------|------------------------------|
| <b>UN-No</b>                | UN2811                       |
| <b>Proper Shipping Name</b> | Toxic solid, organic, n.o.s. |
| <b>Hazard Class</b>         | 6.1                          |
| <b>Packing Group</b>        | III                          |

### IATA

|                             |                              |
|-----------------------------|------------------------------|
| <b>UN-No</b>                | UN2811                       |
| <b>Proper Shipping Name</b> | Toxic solid, organic, n.o.s. |
| <b>Hazard Class</b>         | 6.1                          |
| <b>Packing Group</b>        | III                          |

### IMDG/IMO

|                             |                              |
|-----------------------------|------------------------------|
| <b>UN-No</b>                | UN2811                       |
| <b>Proper Shipping Name</b> | Toxic solid, organic, n.o.s. |
| <b>Hazard Class</b>         | 6.1                          |
| <b>Packing Group</b>        | III                          |

## 15. Regulatory information

### International Inventories

| Component               | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-------------------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Benzenamine, 4-ethenyl- | 1520-21-4 | -   | X    | X    | ACTIVE                                        | 216-185-8 | -      | -   |

| Component               | CAS-No    | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-------------------------|-----------|-------|------|------|------|------|------|-------|-------|
| Benzenamine, 4-ethenyl- | 1520-21-4 | -     | -    | -    | -    | X    | -    | -     | -     |

### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory  
 EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
 IECSC - Chinese Inventory of Existing Chemical Substances  
 KECL - Korean Existing and Evaluated Chemical Substances  
 ENCS - Japanese Existing and New Chemical Substances  
 AICS - Australian Inventory of Chemical Substances  
 PICCS - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Other International Regulations

Authorisation/Restrictions according to EU REACH Not applicable

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component               | CAS-No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Benzenamine, 4-ethenyl- | 1520-21-4 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component               | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Benzenamine, 4-ethenyl- | 1520-21-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

### 16. Other information

**Prepared By** Product Safety Department  
 Email: chem.techinfo@thermofisher.com  
 www.thermofisher.com

**Creation Date** 13-August-2018  
**Revision Date** 30-March-2024  
**Print Date** 30-March-2024  
**Revision Summary** New emergency telephone response service provider.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**